Do Walmart Supercenters Improve Food Security?

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Abstract

This paper examines the effect of Walmart Supercenters, which lower food prices and expand food availability, on household and child food insecurity. Our food insecurity-related outcomes come from the 2001-2012 waves of the December Current Population Study Food Security Supplement. Using narrow geographic identifiers available in the restricted version of these data, we compute the distance between each household's census tract of residence and the nearest Walmart Supercenter. We estimate instrumental variables models that leverage the predictable geographic expansion patterns of Walmart Supercenters outward from Walmart's corporate headquarters. Results suggest that closer proximity to a Walmart Supercenter improves the food security of households and children, as measured by number of affirmative responses to a food insecurity questionnaire and an indicator for food insecurity. The effects are largest among low-income households and children, but are also sizeable for middle-income children.

Keyword*:* Walmart, Wal-Mart, supercenters, big box, food insecurity, hunger

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I. Introduction

The United States Department of Agriculture's (USDA's) Food and Nutrition Service has long sought to end hunger. To this end, they have worked to establish programs aimed at providing ready access to cheap and nutritious foods. Meanwhile, some states and municipalities (like Los Angeles, Chicago, and New York City) have passed laws, taxes, and mandates seeking to block or restrict entry by Walmart, America's largest corporation and largest grocer.[1] Research has shown that Walmart Supercenters lower food prices and increase food availability,[2] so we ask, so we ask: are barriers to Walmart's entry at odds with the goal of eliminating hunger? We add to existing research on food security by exploring the relationship between the diffusion of new retail technologies—specifically, the Walmart Supercenter mass merchandiser format—and various measures of food insecurity.

The United States Department of Agriculture defines food security as "access by all household members at all times to enough food for an active, healthy life."[3] 14.5% of U.S. households lacked food security during at least some of the year 2012. 5.7% had "very low food security", meaning at least one household member had reduced food intake and disrupted eating patterns due to a lack of money or other resources for food (Coleman-Jensen et al. 2013). These rates are particularly striking in light of American wealth and the existence of a federal program – the Supplemental Nutrition Assistance Program – that specifically aims to provide every U.S. household with sufficient resources for a balanced diet.

Food insecurity has been linked to adverse health outcomes. Among adults, these include poor nutrient intakes (McIntyre et al. 2003; Kirkpatrick and Tarusak 2007), mental health

[1] Even when the company is not mentioned by name, some of the proposals are written in such a way as to only apply to Walmart (Hicks 2007, pp. 267-293).
[2] See Basker and Noel (2009), Hausman and Leibtag (2009), Matsa (2011), and Courtemanche and Carden (2014).
[3] See https://www.ers.usda.gov/topics/food-nutrition-assistance/food-security-in-the-us/measurement.aspx.

problems (Heflin et al. 2005), decreased physical health (Tarasuk 2001), depression (Whitaker et al. 2006), diabetes (Seligman et al. 2007), higher prevalence of chronic disease (Seligman et al. 2010), and worse outcomes on health exams (Stuff et al. 2004). Food insecurity among children is associated with higher risks of certain birth defects (Carmichael et al. 2007), anemia (Skalicky et al. 2006; Eicher-Miller et al. 2009), decreased nutrient intakes (Cook et al. 2004), cognitive difficulties (Howard 2011), greater aggression and anxiety (Whitaker 2006), asthma (Kirkpatrick et al. 2010), behavioral problems (Huang et al. 2010), oral health problems (Muirhead et al. 2009), and poorer general health (Cook et al. 2006). Although it is typically assessed by subjective responses to a series of closed end questions, these studies make clear that food insecurity is correlated with many important aspects of health and well-being. Moreover, understanding the relationship between food insecurity and health outcomes is particularly important for researchers interested in low-income families, for whom medical treatment is a luxury good and for whom, therefore, the relative returns to prevention are higher (Steckel 2008, pp. 136-137).

Debate continues about the causes of food insecurity and the roles for the public and private sectors in solving the problem. Individual factors linked to food insecurity in the U.S. include economic variables such as low wages, poverty, unemployment, low education levels, high housing costs, renting rather than owning one's home, and high medical expenses. They also include demographic characteristics such as having children in the home and being African American or Hispanic, unmarried, and relatively young (Ziliak, et al. 2008; City Policy Associates 2010; Gundersen et al. 2011b; Gundersen and Ziliak 2014).

Evidence suggests that SNAP and local food assistance programs ameliorate food insecurity to at least some extent, but concerns about access to these programs remain for some

households while others remain food insecure despite receiving SNAP benefits (Gundersen et al. 2011a; Gregory et al. 2013). The limited effectiveness of public programs has drawn attention to the role of the local food environment in perpetuating food insecurity. In particular, researchers have shown that high food prices and limited access to food stores are both associated with food insecurity (Gregory and Coleman-Jensen 2011; Bonanno 2015). However, the extent to which these associations reflect causal effects is unclear.

We advance the discussion about the relationship between the local food environment and food security by examining how a broad structural change in the American grocery sector – the rapid expansion of Walmart's Supercenter model – affected food security. In 1988, Walmart expanded into large-scale food retail when it opened its first Walmart Supercenter, and the company grew into the U.S.' largest grocer with a 19% market share (Hayes 2013). Walmart Supercenters are by far the dominant supercenter chain, with 3561 locations in 2018 compared to just 239 Super Targets.[4] In the last decade, scholars have turned their attention to the effects of Walmart entry and changes in the structure of the grocery industry on health outcomes (Courtemanche and Carden 2011; Michelson et al. 2018; Borrescio-Higa 2015).[5] Courtemanche and Carden (2011) find that Walmart Supercenters increase obesity, while Volpe et al. (2013) find supercenter entry is followed by a larger proportion of food consumption that is "unhealthy." However, these studies did not examine the other side of the coin of Walmart's "Everyday Low Prices" and expanded food availability: Walmart could also improve food

[4] These numbers come from https://www.statista.com/statistics/269425/total-number-of-walmart-stores-in-the-united-states-by-type/ and https://www.statisticbrain.com/target-company-statistics/.

[5] More generally, Walmart *per se* has become the subject of a ponderous body of research analyzing its effects on everything from (for example) social capital, values, and leisure activities (Goetz and Rupasingha 2006; Carden et al. 2009a, 2009b; Carden and Courtemanche 2009) to the size of the tax base and tax rates (Vandegrift and Loyer 2015; Vandegrift 2016) and even real estate prices (Caceres and Geoghegan 2017; Pope and Pope 2015). See Carden (2013) and Carden and Courtemanche (2016) for summaries of the literature on the mass-market merchandise sector.

security, which is primarily a measure of the quantity of food available to a household, as opposed to its nutritional quality.[6]

Walmart Supercenter entry can provide a significant shock to an area's food availability and food prices. Walmart often opens stores in or near "food deserts" that lack sufficient options for fresh produce. Matsa (2011) shows that Walmart entry leads to better inventory control and fewer stock-outs by competitors, which should improve the reliability of food access. Basker and Noel (2009) and Courtemanche and Carden (2014) show that Walmart Supercenters reduce local food prices both by underselling existing food retailers and by inducing these competitors to lower their own prices. Basker and Noel (2009) also find that the largest price reductions after Walmart Supercenter entry come at stores that serve primarily low-income consumers. Hausman and Leibtag (2009) and Furman (2005) argue that the consumer benefits from diffusion of mass-market merchandisers are considerable and progressively distributed: the major beneficiaries of these firms' lower prices are low-income consumers who spend large percentages of their incomes on food.[7] Hwang and Park (2016) study the effect of Supercenter conversion on shoppers' behavior and find that the conversion leads to larger per-visit expenditures, suggesting that Supercenters affect the local food environment in part by inducing changes in shopping technologies.

We estimate the impacts of Walmart Supercenters on food security using data from the 2001-2012 waves of the December Current Population Study Food Security Supplement (CPS-FSS). Narrow geographic identifiers available in the restricted version of these data enable us to

[6] As we show below, only two out of the 18 questions in the food security module address anything related to diet quality. The rest are aimed at establishing the quantity of food available to the household.

[7] Additionally, the income effect from Walmart's effect on non-food prices could lead to more spending on food and better food security. In a back-of-the-envelope calculation, Courtemanche and Carden (2011) argue that Walmart Supercenters' direct and indirect price effects saved the average household $177 in 2002—additional income that could be spent on more or better nutrients.

compute the distance from each household's census tract to the nearest Walmart Supercenter. Our outcomes are counts of the number of affirmative responses on the household and child-specific portions of the food insecurity questionnaire along with binary variables for household food insecurity, household very low food security, child food insecurity, and child very low food security. We estimate instrumental variables (IV) models that leverage the predictable geographic expansion patterns of Walmart Supercenters outward from corporate headquarters. Specifically, we instrument for Walmart Supercenters with the interaction of distance from Bentonville, Arkansas (Walmart's headquarters) with time. For both households in general and children, the results show that closer proximity to the nearest Walmart Supercenter leads to sizeable and statistically significant improvements in all food security measures except the indicator for very low food security. Subsample analyses reveal that the effects are especially large for low-income households and children, though they are also sizeable for middle-income children.

II. Methods

We begin by estimating linear probability models (LPMs) of the form

$$Y_{ict} = \beta_0 + \beta_1 \ln(DIS_WS_{ct}) + \sum_{j=1}^{J} \gamma_j X_{jict} + \sum_{y=1}^{Y} \tau_y YR_y + \varepsilon_{ict} \qquad (1)$$

where Y_{ict} is the outcome (each of the aforementioned food insecurity variables) for household *i* living in census tract *c* in year *t*, DIS_WS_{ct} is distance in miles from census tract *c* to the nearest Walmart Supercenter in year *t*, X_{jict} is a set of *J* control variables, YR_y is a set of *Y* year fixed effects ($YR_y = 1$ if $y = t$), ε_{ict} is the error term, and the other Greek letters are parameters to be

estimated.[8] Distance from a census tract to the nearest Walmart Supercenter indicates to what extent residents are exposed to Walmart Supercenters, and therefore β_1 measures the association between Walmart Supercenters and households' food insecurity. We take the natural logarithm of distance since it seems reasonable to expect a diminishing marginal effect. For instance, if a new Walmart Supercenter reduces a household's distance to the nearest Walmart Supercenter from 50 to 40 miles, this is unlikely to matter since both stores are prohibitively far away. Standard errors are heteroskedasticity-robust and clustered by census tract, since census tract is the geographic level at which we measure Walmart Supercenter exposure.

Supercenter locations are likely endogenous, which complicates our interpretation of equation (1). First, omitted variable bias could result if changes over time in unobserved area characteristics influence both the entry of Walmart Supercenters and residents' levels of food security. We are able to control for some obvious confounders such as income, but it is difficult to account for all of them. Second, results may be driven by reverse causality as big box grocers may specifically target areas with limited food supply. Indeed, *Dukes v. Wal-Mart* lead plaintiff Betty Dukes criticized Wal-Mart for "promoting themselves to low-income people…They don't put Wal-Mart in those high-end parts of the community. They plant themselves right in the middle of Poorville" (Featherstone 2005).

We address these endogeneity concerns with instrumental variables, or variables that are strongly correlated with the endogenous store variables but otherwise uncorrelated with the outcome (food insecurity) variables conditional on the controls. We adopt a similar strategy used by Courtemanche and Carden (2011) to investigate the impact of Walmart Supercenters on

[8] In unreported regressions, we verified that the estimated marginal effects are virtually identical using probit and logit models instead of linear probability models. This is not surprising since LPMs have been shown to give reliable estimates of average effects (e.g. Angrist and Pischke 2008, Section 3.4.2). We prefer to focus on the LPM estimates since they are easier to implement in the subsequent instrumental variables regressions.

obesity. This strategy is based on the observation that the pattern of Walmart Supercenter expansion starting in 1988 was to radiate outward from the area surrounding Walmart's headquarters in Bentonville, AR. In other words, in the first few years, areas close to northwest Arkansas were the most likely to experience Walmart Supercenter entry, then as time passed entry became more likely in areas progressively further away. Distance from Bentonville therefore influenced the probability an area experienced Walmart Supercenter entry in a given year, and this effect changed over time. This implies that the interaction of distance from Bentonville with time provides plausibly exogenous variation that can identify the causal impact of Walmart Supercenters on food security.[9] Our sample period will be 2001-2012, meaning that Walmart's Supercenter expansion pattern had already been underway for over a decade before our sample starts. Figures 1 through 6 show that the most significant store opening activity during this period was in major metropolitan areas relatively far from Bentonville, implying a diminishing relationship over time between distance from Bentonville and distance from Walmart.

We operationalize the IV strategy by dividing the U.S. into 17 distance rings reflecting 100-mile increments of distance from Bentonville (e.g. less than 100 miles, 100-200 miles, …, 1600 or more miles) and creating an indicator variable for each ring.[10] The distance ring dummies are included as controls, while the interactions of the distance ring dummies with (linear) year are used as instruments. Additionally, to address the possible concern that coastal and inland areas may have experienced different labor market shocks (and therefore different shocks to food security) during the Great Recession and subsequent recovery, we also control for

[9] This strategy is modeled after those of Neumark et al. (2008) and Dube et al. (2007), who observed a similar pattern for Walmart discount stores (as opposed to Supercenters) starting in the 1960s and used it to study the effect of Walmart on local labor markets.

[10] The 100-mile distance ring classification follows Neumark et al. (2008) and Dube et al. (2007).

the interaction of each distance ring with the national unemployment rate.[11] The resulting two-stage IV model therefore has the first-stage equation

$$\ln(DIS_WS_{ct}) = \delta_0 + \sum_{j=1}^{J} \theta_j X_{jict} + \sum_{y=1}^{Y} \rho_y YR_y + \sum_{d=1}^{D} \sigma_d DIS_BEN_d$$

$$+ \sum_{d=1}^{D} \varphi_d \left(DIS_BEN_d * YR_y\right) + \sum_{d=1}^{D} \omega_d \left(DIS_BEN_d * UNEM_y\right) + \mu_{ict} \quad (2)$$

where DIS_BEN_d is the distance from census tract c to Bentonville, $UNEM_y$ is the national unemployment rate, the other variables are the same as in (1), μ_{ict} is the error term, and the other Greek letters are coefficients. The second-stage regression is

$$Y_{ict} = \beta_0 + \beta_1 \ln(\widehat{DIS_{WS}}_{ct}) + \sum_{j=1}^{J} \gamma_j X_{jict} + \sum_{y=1}^{Y} \tau_y YR_y + \sum_{d=1}^{D} \alpha_d DIS_{BEN\,d}$$

$$+ \sum_{d=1}^{D} w_d \left(DIS_BEN_d * UNEM_y\right) + \varepsilon_{ict} \quad (3)$$

This differs from the naïve regression (1) by replacing the distance to Walmart Supercenters with the predicted values of this variable estimated in equation (2) and adding the distance ring fixed effects and their interactions with national unemployment rate as controls.

Identification of β_1 in the IV model comes from the assumption that the distance*year interactions can be excluded from the second-state regression (3) – i.e. that these interactions are uncorrelated with changes over time in food security conditional on the controls. By including

[11] Basker (2005) criticizes Neumark et al. (2008) and Dube et al.'s (2007) distance-from-Bentonville-based identification strategy on the grounds that areas close to Bentonville (inland areas) experience differential labor market fluctuations than those far from Bentonville (coastal areas). Courtemanche and Carden (2011) conduct a wide array of robustness checks and placebo tests to verify that Basker's criticism did not apply to at least one health-related context (obesity). Nonetheless, we find that omitting the distance ring*unemployment rate interactions does meaningfully influence the estimated effects on food security, so we include them in all specifications.

the distance ring fixed effects in (3), we allow for the distances to be correlated with *levels* of food security; we only need to assume that they are uncorrelated with *trends*. We will later assess the validity of the identifying assumption through a number of robustness checks.

III. Data

Our individual-level data come from the 2001-2012 waves of the Current Population Survey Food Security Supplement, an annual household survey conducted by the U.S. Census Bureau for the USDA. The CPS-FSS is the December supplement to the monthly Current Population Survey (CPS), which is a nationally representative survey on labor force statistics. The participants of the CPS-FSS are the same as those interviewed by the original monthly CPS. In the month when the CPS-FSS is conducted, the participants answer the labor force questions as well as a series of questions concerning food security, food consumption, and the usage of food assistance programs.

The CPS-FSS includes the standard set of 18 questions that are used to assess household and child food security. These questions are shown in Table 1. We use responses to this questionnaire to construct four dependent variables for both households in general and children in those households specifically. The first is a simple count of affirmative responses, where the value ranges from zero to eighteen (ten) for households with (without) children, and zero to eight for children (since only the last eight questions focus on children). The remaining outcomes are dichotomized variables commonly used in the literature (e.g. Nord et al. 2005). “Household food insecurity” is defined as three or more affirmative answers on the questionnaire, while “child food insecurity” indicates two or more affirmative answers on the eight questions pertaining to children. “Household very low food security” is defined as eight or more “yes” answers or six or

more in households without children. “Child very low food security” indicates five or more “yes” answers on the child-specific questions.

We also use the CPS-FSS to construct three sets of individual-level control variables: demographic characteristics, economic characteristics, and participation in government food assistance programs. The demographic variables include adult responder’s age; number of own children (dummies for 1, 2, 3, 4, and 5+, with 0 as the omitted base category); dummies for whether race/ethnicity is non-Hispanic white, non-Hispanic black, or Hispanic (with other race/ethnicity as the omitted category); dummies for married and formerly married (with never married as the omitted category); and dummies for high school degree but no further, some college, college degree, and graduate degree (with less than high school degree as the omitted category). The economic variables are household income (dummies for the 16 categories given by the CPS), occupation (dummies for 17 categories), and median income in the census tract.[12] The food assistance variables are indicators for whether any household member received Supplemental Nutrition Assistance Program (SNAP) benefits; Women, Infants, and Children (WIC) benefits; and free/reduced-price school breakfasts or lunches in the past year.

Our independent variable of interest is distance from each respondent’s census tract of residence to the nearest Walmart Supercenter as of the end of each year. Walmart Supercenter entry dates and locations were pieced together from several sources. We began with a list of Walmart Supercenter addresses as of July 2014 from the Brigades Open Data Network.[13]

[12] About 15% of the sample has missing income data. We drop these individuals for the reported regressions, but the results are very similar if we include them and indicate them with a dummy variable.

[13] The list, which includes all Walmart discount stores, Supercenters, and Neighborhood Markets as well as Sam’s Clubs in both the United States and Canada, is available at https://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=44&ved=0CC8QFjADOCg&url=https%3A%2F%2Fbrigades.opendatanetwork.com%2Fapi%2Fviews%2F5gyf-irpw%2Frows.pdf%3Fapp_token%3DU29jcmF0YS0td2VraWNrYXNz0&ei=RKq5U5bdB4qayASlpoC4CQ&usg=AFQjCNH42BpWYSF7E4o7DXLhsbOrIL1_uA (accessed as recently as June 2018).

However, this list does not include the dates on which the stores opened. Entry dates through 2003 come from Thomas Holmes' website and were used in Holmes (2011).[14] We previously updated these data through 2007 using press releases on store openings (either new stores or conversions from discount stores to Supercenters) from Walmart News; these data were used in Courtemanche and Carden (2011, 2014). After 2007, the company only issued online press releases for some store openings and supercenter conversions, leaving 307 Supercenters (out of 3294) unmatched to a more specific entry date. We deduced the year of entry for these remaining stores using a combination of: 1) annual zip code- and county-level counts of supercenters/warehouse clubs from the U.S. Census Bureau's County Business Patterns, 2) the date of the store's first Yelp review that indicates it is a Supercenter, 3) phone calls to the stores, and 4) searching of Internet news articles.[15]

The census tract geographic data come from the U.S. Census Bureau Tiger/Line Shapefiles 2000. For each census tract, an internal point, usually a geographic center of the area, is identified, and its latitude and longitude coordinates are used to label the census tract. The distance from a census tract to the nearest Walmart Supercenter is then computed using the geodetic distance between the two sets of corresponding coordinates. The geodetic distance measures the length of the shortest arc between two points on the ellipsoid surface of the earth. The distance from a census tract to the headquarters of Walmart at Bentonville, AR, is calculated following the same algorithm, then categorized into 17 distance rings.

[14] Holmes' data are available at http://users.econ.umn.edu/~holmes/data/WalMart/.

[15] The zip-code level files appear to classify all supercenters/warehouse clubs that opened between 2008 and 2012 as opening in 2012. This means that those files are only useful for identifying which stores opened after 2012, i.e. after our sample period. The county-level files identify exact years. However, in counties where supercenters/warehouse clubs opened in multiple years between 2008 and 2012, it is not possible to tell which year pertains to the store in question, necessitating the other data collection strategies. Yelp reviews proved particularly useful, as very often a reviewer comments soon after a Supercenter opens to talk about the new (or expanded) store.

We also include a set of controls for proximity to other discount grocers constructed using a similar method. These controls are distance to the nearest Walmart Neighborhood Market, Sam's Club, and Costco. The Neighborhood Market is Walmart's newest store format, and it is similar in size and scope to a traditional grocery store. Sam's Club, owned by Walmart, and Costco are the two leading warehouse club chains. We collected data on Neighborhood Market and Sam's Club entry dates and locations using the same process described for Supercenters above. Costco lists each store location and opening date on its website.

Merging the CPS-FSS to the census tract-level geographic data and county-level store variables requires precise geographic identifiers that are not available in the public-use data. We therefore use the restricted version of the CPS-FSS, provided by the Census Bureau after an application process and accessed in the Atlanta Census Research Data Center.

Dropping observations with missing data yields a final analysis sample of approximately 396,000 households, 120,000 of which have children. Following Research Data Center disclosure policies, we are only able to report sample sizes rounded to the nearest 1,000 observations. Table 2 presents the summary statistics, both for the full sample of all households and the subsample of households with children, for the food security variables as well as distances from the nearest Walmart Supercenter and Bentonville. 14% of households are food insecure, with 5% having very low food security. Children are food insecure in 10% of households, though the rate of children's very low food security is only 1%. The average household lives 18 miles from the nearest Walmart Supercenter and 844 miles from Bentonville. Appendix Table A1 shows the summary statistics for the control variables while Appendix Table A2 does the same for the dummy variables for distance from Bentonville in 100-mile increments.

IV. Results

Table 3 reports the key results for the baseline OLS and IV regressions for each of the eight food insecurity outcomes. The left half of the table contains the results for the household measures while the right half displays those for the child measures. Coefficient estimates for the independent variable of interest – ln(Distance to the Nearest Walmart Supercenter) – are shown. For the IV regressions, we also report the F statistic from a test of the joint significance of the instruments in the first stage, along with the p-value from the overidentification test of the instruments' validity. The table also contains sample sizes as well as means and standard deviations of the outcome variables, which provide a reference point when interpreting the magnitudes of the coefficient estimates.

The OLS results suggest that Walmart Supercenters, if anything, worsen food insecurity. The coefficient estimates are negative for all eight outcomes, meaning that a greater distance from Walmart is associated with less food insecurity, i.e. better food security. However, only two of the six estimates are statistically significant at the 10% level or better, and their magnitudes are small. For instance, since log-units have a percentage interpretation, a 100% increase in distance from Walmart decreases the count of affirmative household responses by just 0.008, which is just 0.9% of the sample mean for that outcome and 0.4% of a standard deviation. It is unclear, of course, whether these small associations actually reflect detrimental effects of Walmart as opposed to "negative selection" in Walmart locations.

In contrast to the OLS results, the IV estimates imply that the causal effect of Walmart Supercenters is to improve food security. The coefficient estimates are positive for all six outcomes, meaning that greater distance from Walmart worsens food insecurity. These estimates are statistically significant at the 5% level or better for four of the six outcomes. The only

exceptions are the very low food security outcomes, for which the lack of significance could potentially be attributable to the relative lack of variation, as the sample rates are just 5% and 1% for households and children, respectively. Moreover, the magnitudes of the significant effects are, in our view, economically meaningful. For households in general, a 100% increase in distance from Walmart (e.g. from 10 to 20 miles) reduces the number of affirmative responses by 0.047 and the probability of being food insecure by 1.2 percentage points. These magnitudes represent 5.1% and 8.7% of the respective outcomes' sample means and 2.1% and 3.5% of their standard deviations. For the children's food security measures, a 100% increase in distance from Walmart reduces affirmative responses by 0.032 and likelihood of being food insecure by 1.4 percentage points. These magnitudes are 8.7% and 14.3% of the corresponding sample means and 3.4% and 4.7% of the corresponding standard deviations.

The instruments perform reasonably well in the diagnostic tests. The F-statistics are over 100, well beyond the rule-of-thumb critical value of 10, indicating that the instruments are sufficiently strong. The overidentification test results are more difficult to interpret. Conceptually, the overidentification test evaluates whether different subsets of instruments lead to statistically different estimated effects of the endogenous variable, with such differences suggesting that at least one of the instruments is invalid. In our case, the model is only overidentified because of our use of a flexible functional form for distance from Bentonville (separate variables for each 100-mile increment), meaning that it is unclear what a "failed" overidentification test would indicate. Nonetheless, the overidentification test only rejects the null hypothesis at the 5% level in one of the six IV regressions.

V. Robustness Checks

We next conduct a number of robustness checks to address possible critiques of our baseline IV model. First, the large set of control variables could conceivably "control away" part of the causal effect of Walmart. For instance, part of Walmart's effect on food security could potentially operate through its effects on employment and wages, which are theoretically ambiguous and are the subject of debate in the literature (Basker 2005; Cardiff-Hicks et al. 2014; Neumark et al. 2008; Dube and Jacobs 2004; Dube et al. 2007). In this case, including income and employment status could lead to an overcontrolling problem. Similarly, Coleman-Jensen (2011) argues that "nonstandard" work (i.e. irregular hours) is associated with greater food insecurity. If big box chains' influence the share of "nonstandard" versus "standard" work, controlling for occupation could be problematic. Walmart could also conceivably influence food security via participation in nutrition assistance programs such as SNAP or by affecting location decisions of other big box stores, either of which would create overcontrolling issues. Therefore, our first series of robustness checks explores the sensitivity of the results to dropping different subsets of controls. Specifically, we re-estimate the IV model including only demographic controls (dropping the economic, food assistance, and food availability controls), only demographic and economic controls, only demographic and food assistance controls, and only demographic and food availability controls.

The second set of checks goes the other direction, adding further controls to help address remaining concerns about the IV model's identifying assumption that distance from Bentonville is uncorrelated with changes over time in food insecurity. In particular, the densely-populated coastal areas that are relatively far from Bentonville could be trending in a different direction than less-populated inland areas that are relatively close to Bentonville. Our baseline model's detailed array of covariates, which include controls for differential effects of economic cycles

across distance from Bentonville rings, should substantially mitigate this concern. Nonetheless, we conduct two robustness checks that explicitly address differential trends by population and region. The first adds the interaction of year with county population, while the second adds the interaction of year with a dummy variable indicating whether the state is on a coast.

The next concern is that, besides Supercenters, Walmart owns two other types of stores with full grocery departments – Sam's Clubs and Walmart Neighborhood Markets – that could conceivably follow the same geographic rollout pattern, creating a problem for the IV model's exclusion restriction. Recall that our baseline model controls for distance from both of these stores, but does not address the endogeneity of their locations, which could conceivably lead to spillover bias in the Walmart Supercenter coefficient estimator. This concern can be easily dismissed for Sam's Clubs, as Courtemanche and Carden (2011) previously documented that the distance-from-Bentonville expansion pattern for Sam's Clubs was finished before the start of our sample period. In unreported regressions, we confirm this by re-estimating the first-stage of the IV model with distance from Sam's Club as the outcome variable, finding that the F statistics are quite small. Neighborhood Markets are potentially more problematic, though, as a similar analysis reveals that they did indeed follow a distance-from-Bentonville-based expansion pattern during our sample period. We therefore conduct two robustness checks that give Neighborhood Markets "equal treatment" to Supercenters in the empirical model. First, we re-define the Walmart variable to be distance from the nearest Supercenter or Neighborhood Market (whichever is closer). Second, we allow the distance-ring-by-year interactions to instrument for both distance from the nearest Supercenter and distance from the nearest Neighborhood Market (separate variables). Note that we prefer to use these models as robustness checks rather than the main specification since we are unaware of any evidence documenting whether Neighborhood

Markets feature the same price advantages as Supercenters, meaning that their predicted effect on food insecurity is less clear.

The next series of checks experiments with different functional forms for the key variables. The first uses a linear, rather than logarithmic, functional form for the distance from the nearest Supercenter variable. The second uses quadratic distance from Bentonville instead of the series of dummies, meaning that the instruments are distance from Bentonville*year and squared distance from Bentonville*year. The third check allows time to be modeled more flexibly when constructing the instruments; specifically, rather than the instruments being the interaction of each distance ring with linear year, in this model the instruments are a full set of distance-ring-by-year fixed effects.

Finally, we consider an entirely different IV strategy based on distance from the nearest Walmart food distribution center, as measured in 2000, the year before the start of the sample period. The intuition is similar to that behind distance from Bentonville. Early in the sample period, Walmart Supercenters were presumably most likely to open near existing distribution centers, but this relationship should weaken over time as the chain expands and new distribution centers open. In other words, the interaction between proximity to year 2000 distribution center locations and time provides plausibly exogenous variation in Walmart Supercenter entry. Since distribution centers were already scattered across the country by 2000, this approach is less susceptible than the distance-from-Bentonville-based strategy to the criticism of differential shocks across coastal and inland areas. Specifically, we estimate the exact same IV model as before, but replace the 100-mile distance from Bentonville dummy variables with 100-mile distance from food distribution center variables. We also estimate another model in which both

the distance from Bentonville*year and distance from food distribution center*year variables are used as instruments.

Table 4 reports the estimated effects of proximity to Walmart on food security across the different robustness checks. For the four outcomes in which Walmart was statistically significant in the baseline model, there are a total of 52 robustness checks (thirteen for each outcome). Walmart remains significant at the 5% level in all but five of these and the 10% level in all but two. While there is some sensitivity in the magnitudes, we do not view the sensitivity as sufficient to meaningfully affect our conclusions. For households in general, a 100% increase in distance from Bentonville increases the count of affirmative responses by between 0.036 and 0.075 and the probability of being food insecure by to 0.7 to 1.2 percentage points. For children, there is even less sensitivity. The effects range from 0.027 to 0.034 for the count of affirmative responses and 1.2 to 1.4 percentage points for the probability of being food insecure.[16] Moreover, even though the baseline estimate for very low household food security was statistically insignificant, in four of the thirteen robustness checks this effect becomes significant. The effect on very low child food security remains insignificant across all specifications, but this is perhaps not surprising given the very low prevalence of this condition (less than 1%). Interestingly, in the robustness check that separately estimates the effects of Walmart Supercenters and Neighborhood Markets, greater proximity to Neighborhood Markets does not significantly improve any of the food security outcomes, suggesting either that these stores either do not offer the same discounts as Supercenters or cater to a different clientele (e.g. individuals who are not as close to the margin of food insecurity).

[16] We exclude the results using linear distance from the nearest Supercenter from this discussion of magnitudes since the scale of the Supercenter variable becomes different in this specification, meaning the magnitudes are not comparable.

VI. Subsample Analyses

We close our empirical analysis by re-estimating the baseline IV model separately for low, middle, and high income subsamples. Since food insecurity is most prevalent among low-income households, we expect Walmart to lead to the largest improvements for that group. The income cutoffs used for the stratifications are chosen to divide the sample into three groups that are as similar in size as possible. Since the CPS only reports income in ranges, exactly equal sample sizes are not possible. Our low-income subsample includes households with annual incomes below $30,000, while the middle-income subsample includes those with incomes between $30,000 and $59,999 and the high-income group contains those with incomes $60,000 and higher.

Table 5 reports the results. For the low-income group, a 100% increase in distance from the nearest Walmart Supercenter significantly increases a household's count of affirmative responses by 0.108 and probability of being food insecure by a 2.0 percentage points. These magnitudes represent 6% and 7.5% of the corresponding sample means. For children, the increases in number of affirmative responses and probability of being food insecure are 0.05 and 2.7 percentage points, or 6.3% and 12% of the sample means. While the coefficient estimates are larger than those for the full sample presented in Table 3, the effect sizes expressed as a percentage of the sample means are roughly similar. In other words, the reason the effects are relatively large for the low-income subsample is simply because food insecurity is more prevalent among that group.

The bottom two panels of Table 5 display the results for the middle- and high-income subsamples. All estimates are statistically insignificant and relatively small for middle- and high-income households as well as high-income children. Interestingly, though, proximity to Walmart

does appear to influence the food security of middle-income children: a 100% increase in distance raises the number of affirmative responses by 0.044, and the probability of being food insecure by 1.5 percentage points. Expressed as percentages of the sample means, these magnitudes are 14% and 18%. Therefore, while the absolute effects are not as large for middle-income children as they are for low-income children, they are actually larger in percentage terms.

VII. Conclusion

This paper asks whether Walmart Supercenters, which lower food prices and expand food availability, improve food security. We estimate instrumental variables (IV) models that exploit the predictable geographic expansion patterns of Walmart Supercenters outward from corporate headquarters. Our results rely on data from the restricted-access 2001-2012 waves of the December CPS Food Security Supplement, which allow for relatively precise measurement of each household's distance from Walmart. We find that the entry of Walmart Supercenters helps to alleviate food insecurity across most measures for both households and children. The effects are strongest for low-income households and children but are still sizeable for middle-income children.

Our finding contributes to the literature in multiple ways. First, we provide new evidence on the causes of food insecurity. Considerable resources are allocated through food assistance programs toward protecting households, especially children, from food insecurity. However, no research to date has examined the influence of big box grocers on food insecurity. Second, we contribute to the debate about Walmart's health effects. Big box grocers, Walmart Supercenters in particular, are blamed for causing obesity (Courtemanche and Carden 2011; Courtemanche et al. 2015). However, we are the first to study the other side of the coin: how the same cheap and readily available food that drives big box grocers' effect on obesity may also help in fighting

food insecurity. This improvement in food security adds another factor local governments should consider when deciding whether to use policy levers (e.g. taxes, zoning laws) to either encourage or discourage Supercenter entry.

Caveats of our study provide directions for future research. For instance, we focus only on Walmart. Other types of food retailers, such as traditional supermarkets, smaller grocery stores, conveniences stores, and warehouse clubs, could all influence food security in different ways. To our knowledge, researchers have not yet identified quasi-experimental sources of variation to identify the causal effects of these store formats. Additionally, our research does not identify the specific mechanisms through which Walmart Supercenters influence food insecurity. While food prices are perhaps the most likely mechanism, food availability and income effects could play a role as well. Furthermore, the net effect of Walmart on health – encompassing the improvement in food security, the increases in obesity and consumption of unhealthy food found by prior studies, and any effects on other health outcomes that have not yet been documented – remains ambiguous.

Along those lines, in recent years Walmart has launched multiple initiatives to explicitly focus on health. During President Barack Obama's first term Walmart joined a number of other firms and organizations in the Partnership for a Healthier America (Simon et al. 2017). As part of the Partnership, Walmart adopted "Great For You" labeling for items that met the 2010 Dietary Guidelines for Americans, renovated stores where food access was a concern, and worked with suppliers to reformulate some of what they sold in order to reduce sodium, sugar, and other unhealthy additives (Simon et al. 2017). In 2011, Walmart collaborated with Tuskegee University to provide market access for small farmers in Alabama which ultimately led to higher productivity for the farms that were able to participate in the study (Karki et al. 2017). Neither

our identification strategy nor our data – which end in 2012 – are well-suited to identify the impacts of these initiatives, but they could represent an important component of Walmart's health effects moving forward.

References

Angrist, J., and J. Pischke. 2008. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton University Press.

Basker, E. 2005. Job Creation or Destruction? Labor-market Effects of Wal-Mart Expansion. *Review of Economics and Statistics* 87:174-183.

Basker, E., and M. Noel. 2009. The Evolving Food Chain: Competitive Effects of Wal-Mart's Entry Into the Supermarket Industry. *Journal of Economics and Management Strategy* 18:977-1009.

Bonanno, A. 2015. Food Insecurity and Food Access in U.S. Metropolitan Areas. *Applied Economic Perspectives and Policy* 37:177-204.

Borrescio-Higa, F. 2015. Can Walmart Make Us Healthier? Prescription Drug Prices and Health Care Utilization. *Journal of Health Economics* 44:37-53.

Caceres, B.C., and J. Geoghegan. 2017. Effects of New Grocery Store Development on Inner-City Neighborhood Residential Prices. *Agricultural and Resource Economics Review* 46(1):87-102.

Carden, A. 2013. Retail Innovations in American Economic History: The Rise of Mass-market Merchandisers. In *The Handbook of Major Events in Economic History,* ed. Randall Parker and Robert Whaples, 402-414. London: Routledge.

Carden, A., and C. Courtemanche. 2009. Wal-Mart, Leisure, and Culture. *Contemporary Economic Policy* 27(4):450-461.

Carden, A., C. Courtemanche, and J. Meiners. 2009a. Does Wal-Mart Reduce Social Capital? *Public Choice* 138:109-136.

Carden, A., Courtemanche, C., and J. Meiners. 2009b. Painting the Town Red? Wal-Mart and Values. *Business and Politics* 11(2):Article 5.

Carden, A., and C. Courtemanche. 2016. The Evolution and Impact of the General Merchandise Sector. In *Handbook of the Economics of Retailing and Distribution,* ed. Emek Basker, 413-432. UK: Edward Elgar.

Cardiff-Hicks, B., F. Lafontaine, and K. Shaw. 2015. Do Large Modern Retailers Pay Premium Wages? *ILR Review* 68(3):633-665.

Carmichael, S.L., W. Yang , A. Herring , B. Abrams, and G. Shaw. 2007. Maternal Food Insecurity Is Associated with Increased Risk of Certain Birth Defects. *Journal of Nutrition* 137:2087–2092.

City Policy Associates. 2010. Status Report on Hunger and Homelessness. Report prepared for Ref: U.S. Conference of Mayors. Washington, DC.

Coleman-Jensen, A.J. 2011. Working for Peanuts: Nonstandard Work and Food Insecurity Across Household Structure. *Journal of Family and Economic Issues* 32:84-97.

Coleman-Jensen, A., M. Nord, and A. Singh. 2013. Household Food Security in the United States in 2012. ERR-155, U.S. Department of Agriculture, Economic Research Service.

Cook, J., D. Frank, C. Berkowitz, M. Black, P. Casey, D. Cutts, A. Meyers, N. Zaldivar, A. Skalicky, S. Levenson, and T. Heeren. 2004. Food Insecurity Is Associated with Adverse Health Outcomes Among Human Infants and Toddlers. *Journal of Nutrition* 134:1348–1432.

Cook, J., D. Frank, S. Levenson, N. Neault, T. Heeren, M. Black, C. Berkowitz, P. Casey, A. Meyers, D. Cutts, and M. Chilton. 2006. Child Food Insecurity Increases Risks Posed by Household Food Insecurity to Young Children's Health. *Journal of Nutrition* 136:1073–1076.

Courtemanche, C., and A. Carden. 2011. The Impact of Walmart Supercenters on Body Mass Index and Obesity. *Journal of Urban Economics* 69:165-181.

Courtemanche, C., and A. Carden. 2014. Competing with Costco and Sam's Club: Warehouse club Entry and Grocery Prices. *Southern Economic Journal* 80:565-585.

Courtemanche, C., J. Pinkston, C. Ruhm, and G. Wehby. 2015. Can Changing Economic Incentives Explain the Rise in Obesity? Working Paper 20892, National Bureau of Economic Research.

Dube, A., and K. Jacobs. 2004. Hidden Cost of Wal-Mart jobs: Use of Safety Net Programs by Wal-Mart Workers in California. Center for Labor Research and Education, Institute of Industrial Relations, University of California.

Dube, A., T. Lester, and B. Eidlin. 2007. Firm Entry and Wages: Impact of Wal-Mart Growth on Earnings Throughout the Retail Sector. Working Paper iirwps-126-05, Institute of Industrial Relations.

Eicher-Miller, H.A., A. Mason, C. Weaver, G. McCabe, and C. Boushey. 2009. Food Insecurity Is Associated with Iron Deficiency Anemia in U.S. Adolescents. *American Journal of Clinical Nutrition* 90:1358–1371.

Featherstone, L. 2005. Down and Out in Discount America. *The Nation* 3:11-15.

Furman, J. 2005. Wal-Mart: A Progressive Success Story. Center for American Progress.

Goetz, S. J., and A. Rupasingha. 2006. Wal-Mart and Social Capital. *American Journal of Agricultural Economics* 88:1304-1310.

Gregory, C., and A. Coleman-Jensen. 2011. Do Food Prices Affect Food Security? Evidence from the CPS 2002-2006. Paper presented at the Agricultural and Applied Economics Association Annual Meeting, Pittsburgh, PA.

Gregory, C., M. Rabbitt, and D. Ribar. 2013. The Supplemental Nutrition Assistance Program and Food Insecurity. *SNAP matters: How food stamps affect health and well-being*, 74-106.

Gundersen C, and J.P. Ziliak. 2014. Childhood Food Insecurity in the U.S.: Trends, Causes, and Policy Options. *The Future of Children* 24(2):1-19.

Gundersen, C., B. Kreider, and J. Pepper. 2011a. Partially Identifying the Impact of the Supplemental Nutrition Assistance Program on Food Insecurity Among Children: Addressing Endogeneity and Misreporting Using the SIPP. National Poverty Center Working Paper Series.

Gundersen C, B. Kreider, and J. Pepper. 2011b. The Economics of Food Insecurity in the United States. Applied Economic Perspectives and Policy 33(3):281–303.

Hausman, J., and E. Leibtag. 2009. Consumer Benefits from Increased Competition in Shopping Outlets: Measuring the Effect of Wal-Mart. *Journal of Applied Econometrics* 22:1157-1177.

Hayes, F. 2013. U.S. Supermarkets Grab Market Share From Walmart And Warehouse Chains. Available http://www.fierceretail.com/story/us-supermarkets-grab-market-share-walmart-and-warehouse-chains/2013-06-02.

Heflin, C.M., K. Siefert, and D. Williams. 2005. Food Insufficiency and Women's Mental Health: Findings from A 3-Year Panel of Welfare Recipients. *Social Science and Medicine* 61:1971–82.

Hicks, M. 2007. The Local Economic Impact of Wal-Mart. Youngstown, NY: Cambria Press.

Holmes, T. 2011. The Diffusion of Wal-Mart and Economies of Density. *Econometrica* 79:253-301.

Howard, L. 2011. Does Food Insecurity at Home Affect Non-cognitive Performance at School? A Longitudinal Analysis of Elementary Student Classroom Behavior. *Economics of Education Review* 30:157–176.

Huang, J., K. Matta Oshima, and Y. Kim. 2010. Does Food Insecurity Affect Parental Characteristics and Child Behavior? Testing Mediation Effects. *Social Service Review* 84:381–401.

Hwang, M., and S. Park. 2016. The Impact of Walmart Supercenter Conversion on Consumer Shopping Behaviors. *Management Science* 62(3):817-828.

Karki, L.B., W.A. Hill, R. Shange, T.M. Hargrove, and M.D. Robinson. 2017. Assessing the Impact of the Small Farmer-Tuskegee University-Walmart Project on the Household Economy of Small and Limited Resource Farmers in Alabama. *Professional Agricultural Workers Journal* 4(2):1-15.

Kirkpatrick, S., L. McIntyre, and M. Potestio. 2010. Child Hunger and Long-term Adverse Consequences for Health. *Archives of Pediatric Adolescent Medicine* 164:754–762.

Kirkpatrick, S., and V. Tarasuk. 2007. Food Insecurity Is Associated with Nutrient Intakes Among Canadian Adults and Adolescents. *Journal of Nutrition* 138:604–12.

Matsa, D. A. 2011. Competition and Product Quality in the Supermarket Industry. *Quarterly Journal of Economics* 126(3):1539-1591.

McIntyre, L., T. Glanville, K. Raine, J. Dayle, B. Anderson, and N. Battaglia. 2003. Do Low-income Lone Mothers Compromise Their Nutrition to Feed Their Children? *Canadian Medical Association Journal* 168:686–91.

Michelson, H., S. Boucher, X. Cheng, and J. Huang. 2018. Connecting Supermarkets and Farms: The Role of Intermediaries in Walmart China's Fresh Produce Supply Chains. *Renewable Agriculture and Food Systems* 33(1):47-59.

Muirhead, V., C. Quiñonez, R. Figueiredo, D. Locker. 2009. Oral Health Disparities and Food Insecurity in Working Poor Canadians. *Community Dentistry and Oral Epidemiology* 37:294–304.

Nord, M., M. Andrews, and S. Carlson. 2005. Household Food Security in the United States, 2005. United States Department of Agriculture Economic Research Service, Economic Research Report 29.

Neumark, D., J. Zhang, and S. Ciccarella. 2008. The Effects of Wal-Mart on Local Labor Markets. *Journal of Urban Economics* 63:405-430.

Pope, D., and J. Pope. 2015. When Walmart Comes to Town: Always Low Housing prices? Always? *Journal of Urban Economics* 87:1-13.

Seligman, H., A. Bindman, E. Vittinghoff, A. Kanaya, and M. Kushel. 2007. Food Insecurity Is Associated with Diabetes Mellitus: Results from the National Health Examination and Nutritional Examination Survey 1999–2002. *Journal of General Internal Medicine* 22:1018–23.

Seligman, H., B. Laraia, and M. Kushel. 2010. Food Insecurity Is Associated with Chronic Disease Among Low-income NHANES Participants. *Journal of Nutrition* 140:304–10.

Simon, C.S., L. Kocot, and W.H. Dietz. 2017. Partnership for a Healthier America: Creating Change Through Private Sector Partnerships. *Current Obesity Reports* 6(2):108-115.

Skalicky, A., A. Meyers, W. Adams, Z. Yang, J. Cook, and D. Frank. 2006. Child Food Insecurity and Iron Deficiency Anemia in Low-income Infants and Toddlers in the United States. *Maternal and Child Health Journal* 10:177–85.

Steckel, R. H. 2008. Biological Measures of the Standard of Living. *Journal of Economic Perspectives* 22(1):129-152.

Stuff, J.E., P. Casey, K. Szeto, J. Gossett, J. Robbins, P. Simpson, C. Connell, M.L. Bogle. 2004. Household Food Insecurity Is Associated with Adult Health Status. *Journal of Nutrition* 134:2330–2305.

Tarasuk, V.S. 2001. Household Food Insecurity with Hunger Is Associated with Woman's Food Intakes, Health and Household Circumstances. *Journal of Nutrition* 131:2670–2676.

Vandegrift, D. 2016. The Effect of Walmart and Target on Property Tax Rates. *Letters in Spatial and Resource Sciences* 9(3):309-327.

Vandegrift, D., and J. Loyer. 2015. The Effect of Walmart and Target on the Tax Base: Evidence from New Jersey. *Journal of Regional Science* 55(2):159-187.

Volpe, R., A. Okrent, and E. Leibtag. 2013. The Effect of Supercenter-format Stores on the Healthfulness of Consumers' Grocery Purchases. *American Journal of Agricultural Economics*, 95:568-589.

Whitaker, R.C., S. Phillips, and S. Orzol. 2006. Food Insecurity and the Risks of Depression and Anxiety in Mothers and Behavior Problems in Their Preschool-aged Children. *Pediatrics* 118:e859–e868.

Ziliak J.P., C. Gundersen, and M. Haist. 2008. The Causes, Consequences, and Future of Senior Hunger in America. National Foundation to End Senior Hunger.

Figure 1 – Walmart Supercenter Locations in 2002

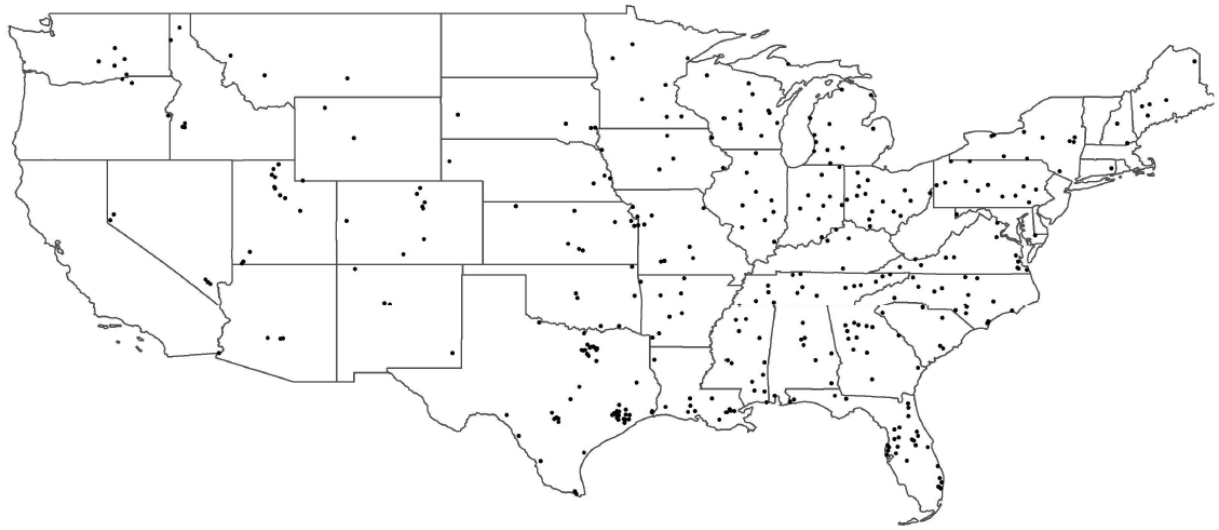


Figure 2 – Walmart Supercenter Locations in 2004

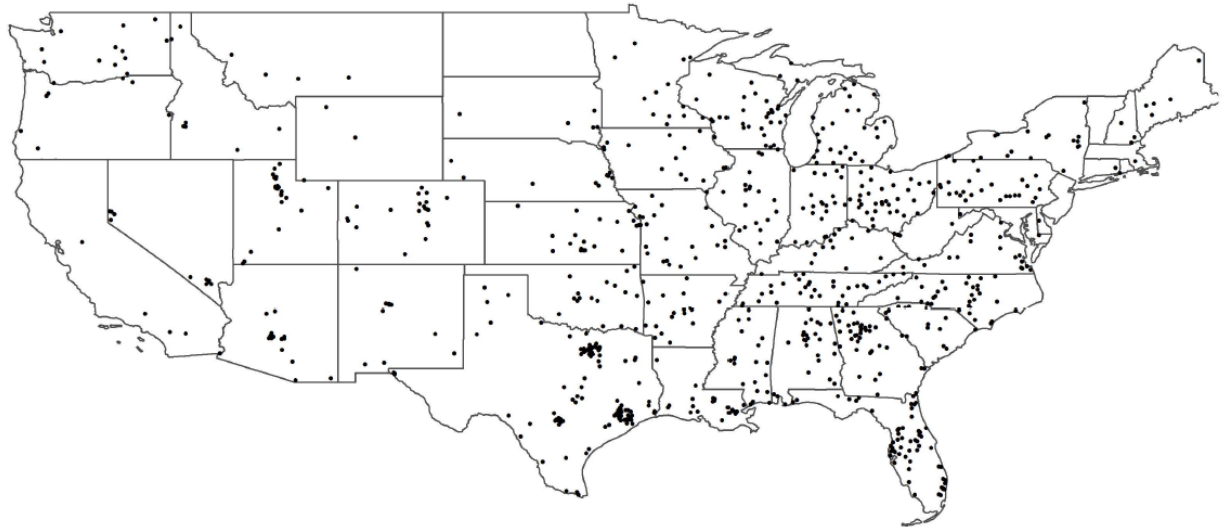


Figure 3 – Walmart Supercenter Locations in 2006

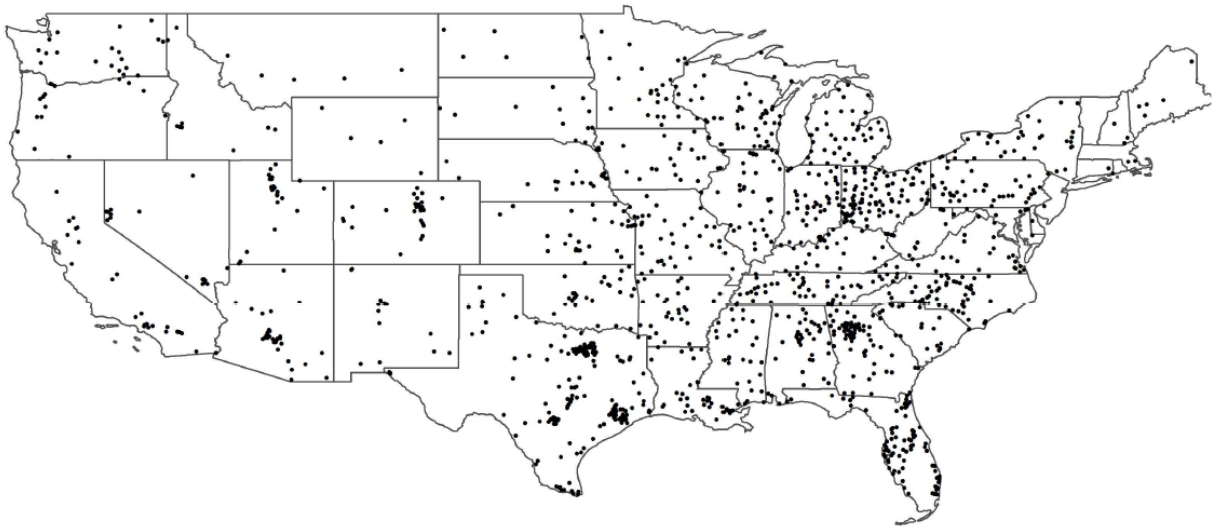


Figure 4 – Walmart Supercenter Locations in 2008

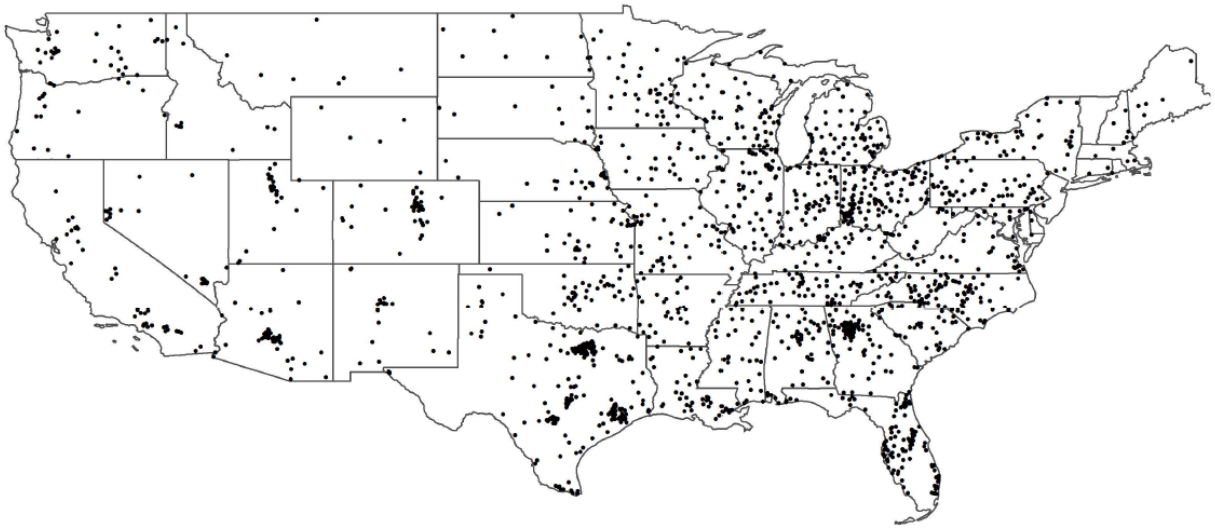


Figure 5 – Walmart Supercenter Locations in 2010

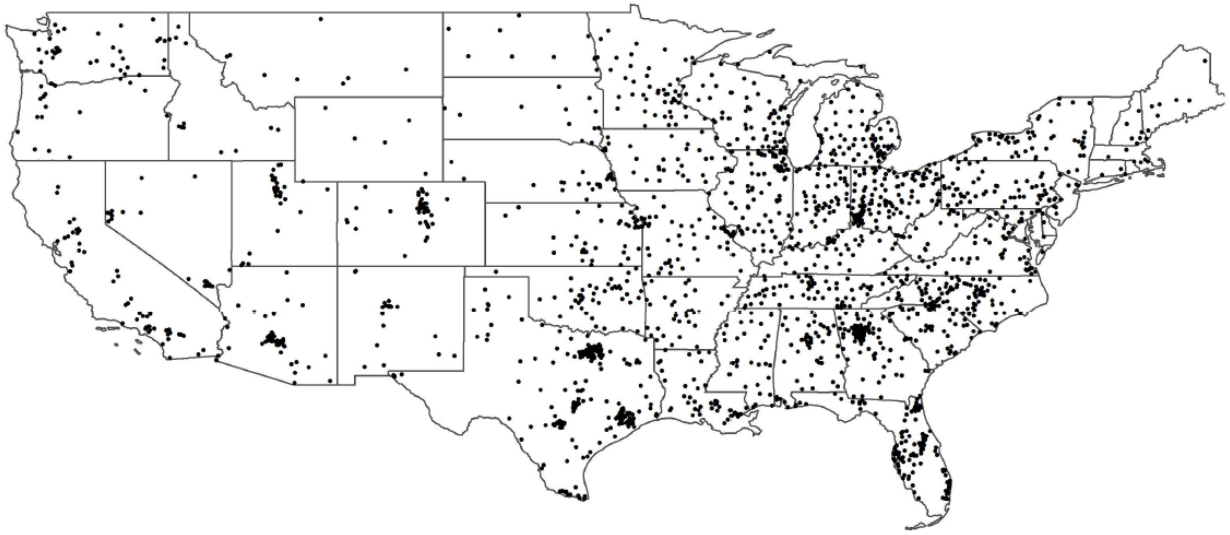


Figure 6 – Walmart Supercenter Locations in 2012

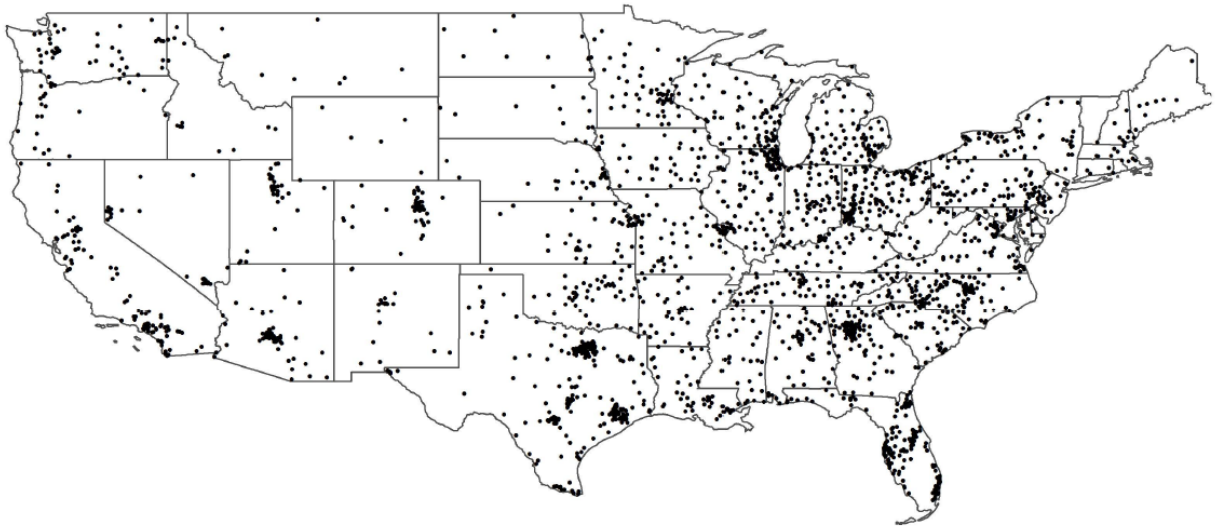


Table 1 – Questions Used To Assess the Food Security of Households in the CPS Food Security Survey

Question Number	Question Text
1	"We worried whether our food would run out before we got money to buy more." Was that often, sometimes, or never true for you in the last 12 months?
2	"The food that we bought just didn't last and we didn't have money to get more." Was that often, sometimes, or never true for you in the last 12 months?
3	"We couldn't afford to eat balanced meals." Was that often, sometimes, or never true for you in the last 12 months?
4	In the last 12 months, did you or other adults in the household ever cut the size of your meals or skip meals because there wasn't enough money for food? (Yes/No)
5	(If yes to question 4) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?
6	In the last 12 months, did you ever eat less than you felt you should because there wasn't enough money for food? (Yes/No)
7	In the last 12 months, were you ever hungry, but didn't eat, because there wasn't enough money for food? (Yes/No)
8	In the last 12 months, did you lose weight because there wasn't enough money for food? (Yes/No)
9	In the last 12 months did you or other adults in your household ever not eat for a whole day because there wasn't enough money for food? (Yes/No)
10	(If yes to question 9) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?
11	"We relied on only a few kinds of low-cost food to feed our children because we were running out of money to buy food." Was that often, sometimes, or never true for you in the last 12 months?
12	"We couldn't feed our children a balanced meal, because we couldn't afford that." Was that often, sometimes, or never true for you in the last 12 months?
13	"The children were not eating enough because we just couldn't afford enough food." Was that often, sometimes, or never true for you in the last 12 months?
14	In the last 12 months, did you ever cut the size of any of the children's meals because there wasn't enough money for food? (Yes/No)
15	In the last 12 months, were the children ever hungry but you just couldn't afford more food? (Yes/No)
16	In the last 12 months, did any of the children ever skip a meal because there wasn't enough money for food? (Yes/No)
17	(If yes to question 16) How often did this happen—almost every month, some months but not every month, or in only 1 or 2 months?
18	In the last 12 months did any of the children ever not eat for a whole day because there wasn't enough money for food? (Yes/No)

Note: Questions 11-18 were asked only if the household included children age 0-17.

Table 2 – Summary Statistics for Key Variables

	All households; household food security	Households with children; children's food security
Count of affirmative household responses	0.925 (2.203)	0.366 (0.931)
Food insecurity	0.138 (0.345)	0.098 (0.298)
Very low food security	0.052 (0.222)	0.009 (0.093)
Distance from nearest Walmart Supercenter	17.69 (29.84)	18.46 (31.31)
Distance from Bentonville, AR	843.8 (391.6)	844.0 (395.1)
Sample Size	396,000	120,000

Notes: Means are shown, with standard deviations in parentheses. CPS household sampling weights are used. Sample sizes are rounded to the nearest 1,000 according to Census Bureau guidelines for restricted data.

Table 3 – Results from Baseline Regressions

	Household			Children		
	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security
Ordinary Least Squares						
ln(Distance to Walmart Supercenter)	-0.008*	-0.0009	-0.0005	-0.005*	-0.001	-0.0003
	(0.004)	(0.0007)	(0.0004)	(0.003)	(0.001)	(0.0003)
Instrumental Variables						
ln(Distance to Walmart Supercenter)	0.047**	0.012***	0.003	0.032**	0.014**	0.0003
	(0.020)	(0.004)	(0.002)	(0.014)	(0.005)	(0.0015)
First Stage F Statistic	139.5	139.5	139.5	102.2	102.2	102.2
Overidentification Test P-Value	0.219	0.096	0.350	0.057	0.034	0.818
Sample Size	396,000	396,000	396,000	120,000	120,000	120,000
Mean (Standard Deviation) of Dependent Variable	0.925	0.138	0.052	0.366	0.098	0.009
	(2.203)	(0.345)	(0.222)	(0.931)	(0.298)	(0.093)

Notes: Standard errors, heteroskedasticity-robust and clustered by census tract, are in parentheses. *** indicates statistically significant at 1% level; ** 5% level; * 10% level. All regressions include demographic, economic, food assistance, and food availability controls as well as year fixed effects. Instrumental variables regressions also include distance ring fixed effects and their interactions with national unemployment rate. CPS household sampling weights are used. Sample sizes are rounded to the nearest 1,000 according to Census Bureau guidelines for restricted data.

Table 4 – Results from Robustness Checks

	Household			Children		
	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security
Fewer Controls						
Demographic controls only	0.048**	0.0085**	0.0034	0.028*	0.0125**	0.0001
	(0.024)	(0.0037)	(0.0023)	(0.016)	(0.0050)	(0.0015)
Demographic and economic controls only	0.061***	0.0105***	0.0041*	0.032**	0.0138***	0.0003
	(0.022)	(0.0034)	(0.0022)	(0.015)	(0.0048)	(0.0015)
Demographic and food assistance controls only	0.052**	0.0090***	0.0035	0.0292**	0.0129***	0.0001
	(0.022)	(0.0034)	(0.0022)	(0.015)	(0.0048)	(0.0015)
Demographic and food availability controls	0.0458	0.0079**	0.0030	0.0314**	0.0137***	0.0002
	(0.0231)	(0.0036)	(0.0022)	(0.0152)	(0.0049)	(0.0015)
Additional Controls						
Add county population*year	0.047**	0.008**	0.003	0.034**	0.014***	0.0004
	(0.020)	(0.003)	(0.002)	(0.014)	(0.005)	(0.002)
Add coastal state*year	0.047**	0.008**	0.003	0.033**	0.014***	0.0004
	(0.020)	(0.003)	(0.002)	(0.014)	(0.005)	(0.001)
Instrument for Neighborhood Market						
ln(Distance to Supercenter or Neighborhood Market)	0.0380*	0.0077**	0.0017	0.0266*	0.013***	-0.0004
	(0.0223)	(0.0035	(0.0023)	(0.0155)	(0.005)	(0.016)
Separate Variables						
ln(Distance to nearest Supercenter)	0.0520**	0.009***	0.003	0.031**	0.013***	0.0005
	(0.0210)	(0.003)	(0.002)	(0.015)	(0.005)	(0.002)
ln(Distance to nearest Neighborhood Market)	-0.0372	-0.004	-0.004	-0.036*	-0.010	-0.004*
	(0.0274)	(0.004)	(0.003)	(0.020)	(0.007)	(0.002)

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	Households			Children		
	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security
Alternate Functional Forms						
Linear distance to nearest Supercenter	0.0009*** (0.0003)	0.0002*** (0.00005)	0.00006* (0.00003)	0.0006*** (0.0002)	0.0002*** (0.0001)	0.00001 (0.00002)
Quadratic distance from Benvonville	0.075*** (0.026)	0.012*** (0.004)	0.007** (0.003)	0.028 (0.018)	0.012** (0.006)	0.002 (0.002)
Distance ring*year fixed effects as instruments	0.036** (0.016)	0.0070*** (0.0026)	0.0021 (0.0015)	0.032*** (0.011)	0.013*** (0.004)	0.0004 (0.0012)
Distance from Nearest Food Distribution Center as Instrument						
Distance from Food Distribution Center Rings*Year as Instrument	0.057*** (0.019)	0.010*** (0.003)	0.0038* (0.0020)	0.034** (0.014)	0.014*** (0.004)	0.0002 (0.015)
Use both distances from Bentonville and Food Distribution Center	0.047*** (0.018)	0.0084*** (0.0029)	0.0025 (0.0018)	0.033*** (0.013)	0.014*** (0.004)	-5.99e-6 (0.0014)

Notes: Standard errors, heteroskedasticity-robust and clustered by census tract, are in parentheses. *** indicates statistically significant at 1% level; ** 5% level; * 10% level. Instrumental variables regressions are shown in all cases. All regressions include demographic, economic, food assistance, and food availability controls; year fixed effects; and distance ring fixed effects and their interactions with national unemployment rate unless otherwise indicated. CPS household sampling weights are used. Sample sizes are 396,000 for all households and 120,000 for households with children, rounded to the nearest 1,000 according to Census Bureau guidelines for restricted data.

Table 5 – Results from Income Stratifications

	Household			Children		
	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security	Count of affirmative responses	Indicator for food insecurity	Indicator for very low food security
Low Income Subsample						
ln(Distance to Walmart Supercenter)	0.108**	0.020***	0.005	0.0504	0.027**	-0.001
	(0.047)	(0.007)	(0.005)	(0.0362)	(0.012)	(0.004)
Sample Size	142,000	142,000	142,000	35,000	35,000	35,000
Mean (Standard Deviation) of Dependent Variable	1.789	0.267	0.110	0.805	0.221	0.022
	(2.887)	(0.442)	(0.313)	(1.295)	(0.415)	(0.147)
Middle Income Subsample						
ln(Distance to Walmart Supercenter)	-0.001	0.002	-0.0007	0.044*	0.015*	0.002
	(0.031)	(0.005)	(0.003)	(0.023)	(0.008)	(0.002)
Sample Size	124,000	124,000	124,000	37,000	37,000	37,000
Mean (Standard Deviation) of Dependent Variable	0.697	0.105	0.032	0.321	0.085	0.006
	(1.857)	(0.306)	(0.176)	(0.828)	(0.279)	(0.074)
High Income Subsample						
ln(Distance to Walmart Supercenter)	0.021	0.003	0.001	0.009	0.004	-0.00006
	(0.015)	(0.003)	(0.001)	(0.009)	(0.003)	(0.0006)
Sample Size	130,000	130,000	130,000	49,000	49,000	49,000
Mean (Standard Deviation) of Dependent Variable	0.1874	0.027	0.007	0.068	0.016	0.001
	(0.9850)	(0.163)	(0.083)	(0.390)	(0.126)	(0.033)

Notes: Standard errors, heteroskedasticity-robust and clustered by census tract, are in parentheses. *** indicates statistically significant at 1% level; ** 5% level; * 10% level. Instrumental variables regressions are shown in all cases. All regressions include demographic, economic, food assistance, and food availability controls; year fixed effects; and distance ring fixed effects and their interactions with national unemployment rate unless otherwise indicated. CPS household sampling weights are used. Sample sizes are rounded to the nearest 1,000 according to Census Bureau guidelines for restricted data.

Appendix Table A1 – Summary Statistics for Control Variables

	All Households; Household Food Security	Households with Children; Children's Food Security
Demographic Controls		
Household is headed by unmarried male	0.206 (0.405)	0.063 (0.242)
Household is headed by unmarried female	0.297 (0.457)	0.242 (0.429)
Age of responder	49.09 (16.99)	38.58 (9.018)
One child under 18 in household	0.132 (0.339)	0.426 (0.494)
Two children under 18 in household	0.115 (0.319)	0.372 (0.483)
Three children under 18 in household	0.045 (0.208)	0.146 (0.353)
Four children under 18 in household	0.013 (0.113)	0.042 (0.2)
Five or more children under 18 in household	0.005 (0.067)	0.015 (0.12)
Live in MSA but outside of central city	0.372 (0.483)	0.393 (0.488)
Live in urban area that is not in MSA	0.107 (0.309)	0.105 (0.307)
Live in rural area	0.217 (0.412)	0.214 (0.41)
Responder is high school graduate but no further	0.293 (0.455)	0.284 (0.451)
Responder has some college but no four-year degree	0.191 (0.393)	0.192 (0.394)
Respondent has college degree but no graduate	0.183 (0.387)	0.188 (0.391)
Respondent has some graduate school	0.104 (0.306)	0.101 (0.301)
Respondent is Hispanic	0.113 (0.317)	0.174 (0.379)
Respondent is non-Hispanic white	0.724 (0.447)	0.646 (0.478)
Respondent is non-Hispanic black	0.115 (0.319)	0.126 (0.332)
Economic Controls		
Respondent's occupation is in agriculture	0.012 (0.109)	0.013 (0.114)
Respondent's occupation is in construction	0.056 (0.23)	0.075 (0.263)
Respondent's occupation is in manufacturing	0.082 (0.275)	0.102 (0.303)
Respondent's occupation is in trade	0.078 (0.268)	0.089 (0.285)
Respondent's occupation is in transportation	0.026 (0.159)	0.03 (0.17)
Respondent's occupation is in utilities	0.007 (0.083)	0.009 (0.092)
Respondent's occupation is in information technology	0.026 (0.158)	0.03 (0.172)
Respondent's occupation is in finance	0.022 (0.145)	0.027 (0.161)
Respondent's occupation is in insurance	0.015 (0.12)	0.019 (0.135)
Respondent's occupation is in health services	0.066 (0.249)	0.085 (0.279)

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	All Households; Household Food Security	Households with Children; Children's Food Security
Respondent's occupation is in educational services	0.057 (0.232)	0.064 (0.246)
Respondent's occupation is in real estate	0.027 (0.161)	0.031 (0.174)
Respondent's occupation is in professional services	0.076 (0.265)	0.089 (0.285)
Respondent's occupation is in other services	0.101 (0.301)	0.124 (0.329)
Respondent's occupation is in public administration	0.033 (0.179)	0.039 (0.193)
Respondent's occupation is in other industry	0.013 (0.114)	0.014 (0.119)
Household income is between $5000 and $7499	0.026 (0.16)	0.022 (0.147)
Household income is between $7500 and $9999	0.031 (0.174)	0.021 (0.144)
Household income is between $10000 and $12499	0.042 (0.2)	0.03 (0.171)
Household income is between $12500 and $14999	0.037 (0.189)	0.03 (0.17)
Household income is between $15000 and $19999	0.057 (0.231)	0.047 (0.211)
Household income is between $20000 and $24999	0.07 (0.254)	0.059 (0.235)
Household income is between $25000 and $29999	0.068 (0.251)	0.061 (0.24)
Household income is between $30000 and $34999	0.068 (0.252)	0.063 (0.243)
Household income is between $35000 and $39999	0.061 (0.289)	0.057 (0.231)
Household income is $40000 and $49999	0.092 (0.289)	0.088 (0.284)
Household income is between $50000 and $59999	0.087 (0.282)	0.089 (0.285)
Household income is between $60000 and $74999	0.096 (0.294)	0.109 (0.312)
Household income is $75000 or more	0.234 (0.423)	0.291 (0.454)
Poverty rate in census tract of residence	0.121 (0.1)	0.12 (0.102)
Food Assistance Controls		
Anyone in household gets free or reduced-price school meals	0.083 (0.276)	0.232 (0.422)
Household receives SNAP benefits	0.088 (0.283)	0.143 (0.351)
Household receives WIC benefits	0.033 (0.178)	0.09 (0.285)
Food Availability Controls		
Distance (in miles) from nearest Walmart Supercenter	17.69 (29.84)	18.46 (31.31)
Distance (in miles) from nearest Sam's Club	20.82 (31.64)	20.55 (31.26)
Distance (in miles) from nearest Costco	46.52 (62.9)	46.36 (63.79)
Distance (in miles) from nearest Neighborhood Market	280.7 (246)	284.3 (249.49)
Distance (in miles) from Bentonville, Arkansas	843.8 (391.6)	844 (395.1)

Notes: Means are shown, with standard deviations in parentheses. CPS household sampling weights are used. Approximate sample sizes are 396,000 for all households and 120,000 for households with children.

Appendix Table A2 – Summary Statistics for Distance from Bentonville Dummy Variables

	All Households; Household Food Security	Households with Children; Children's Food Security
Distance from Bentonville between 100 and 199 Miles	0.026 (0.159)	0.025 (0.157)
Distance from Bentonville between 200 and 299 Miles	0.049 (0.215)	0.051 (0.221)
Distance from Bentonville between 300 and 399 Miles	0.039 (0.194)	0.04 (0.2)
Distance from Bentonville between 400 and 499 Miles	0.089 (0.284)	0.091 (0.288)
Distance from Bentonville between 500 and 599 Miles	0.124 (0.33)	0.125 (0.331)
Distance from Bentonville between 600 and 699 Miles	0.086 (0.28)	0.084 (0.277)
Distance from Bentonville between 700 and 799 Miles	0.08 (0.271)	0.077 (0.267)
Distance from Bentonville between 800 and 899 Miles	0.054 (0.226)	0.048 (0.214)
Distance from Bentonville between 900 and 999 Miles	0.081 (0.273)	0.076 (0.265)
Distance from Bentonville between 1000 and 1099 Miles	0.076 (0.266)	0.076 (0.265)
Distance from Bentonville between 1100 and 1199 Miles	0.084 (0.278)	0.089 (0.28)
Distance from Bentonville between 1200 and 1299 Miles	0.035 (0.185)	0.035 (0.183)
Distance from Bentonville between 1300 and 1399 Miles	0.081 (0.273)	0.091 (0.287)
Distance from Bentonville between 1400 and 1499 Miles	0.024 (0.152)	0.024 (0.154)
Distance from Bentonville between 1500 and 1599 Miles	0.037 (0.188)	0.036 (0.187)
Distance from Bentonville 1600 Miles or Greater	0.027 (0.163)	0.025 (0.157)

Notes: Means are shown, with standard deviations in parentheses. CPS household sampling weights are used. Approximate sample sizes are 396,000 for all households and 120,000 for households with children.

Appendix Table A3 – First-Stage Coefficients of Interest from Baseline Instrumental Variables Regressions

	All Households	Households with Children
Distance from Bentonville 100-199 miles*year	-0.021 (0.017)	-0.0106 (0.0236)
Distance from Bentonville 200-299 miles*year	-0.027 (0.016)*	-0.0050 (0.0214)
Distance from Bentonville 300-399 miles*year	-0.006 (0.016)	0.0041 (0.0221)
Distance from Bentonville 400-499 miles*year	-0.033 (0.015)**	-0.014 (0.0216)
Distance from Bentonville 500-599 miles*year	-0.088 (0.015)***	-0.0725 (0.0209)***
Distance from Bentonville 600-699 miles*year	-0.059 (0.015)***	-0.0484 (0.0210)**
Distance from Bentonville 700-799 miles*year	-0.081 (0.015)***	-0.0666 (0.0216)***
Distance from Bentonville 800-899 miles*year	-0.026 (0.015)	-0.0074 (0.0218)
Distance from Bentonville 900-999 miles*year	-0.014 (0.015)	0.0001 (0.0215)
Distance from Bentonville 1000-1099 miles*year	-0.050 (0.015)***	-0.0314 (0.0214)
Distance from Bentonville 1100-1199 miles*year	-0.026 (0.015)*	0.0007 (0.0209)
Distance from Bentonville 1200-1299 miles*year	-0.058 (0.015)***	-0.0444 (0.0221)**
Distance from Bentonville 1300-1399 miles*year	-0.290 (0.015)***	-0.2977 (0.021)***
Distance from Bentonville 1400-1499 miles*year	-0.173 (0.021)***	-0.1780 (0.0290)***
Distance from Bentonville 1500-1599 miles*year	-0.258 (0.017)***	-0.2527 (0.0229)***
Distance from Bentonville ≥1600 miles*year	-0.257 (0.017)***	-0.2598 (0.0241)***
Sample size	396,000	120,000

Notes: Standard errors, heteroskedasticity-robust and clustered by census tract, are in parentheses. *** indicates statistically significant at 1% level; ** 5% level; * 10% level. The dependent variable is ln(Distance from the Nearest Walmart Supercenter). All regressions include demographic, economic, food assistance, and food availability controls as well as year fixed effects. Instrumental variables regressions also include distance ring fixed effects and their interactions with national unemployment rate. CPS household sampling weights are used. Sample sizes are rounded to the nearest 1,000 according to Census Bureau guidelines for restricted data.